03-09-26

unix_socket_server.c

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main()
{
    int server_fd, client_fd;
    struct sockaddr_un address;
    char buffer[100];

    // Create socket
    server_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    // Set socket address
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    // Remove old socket
    unlink(SOCKET_PATH);

    // Bind socket to path
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));

    // Wait for client
    listen(server_fd, 5);

    printf("Server waiting...\n");

    // Accept client
    client_fd = accept(server_fd, NULL, NULL);

    // Receive message
    read(client_fd, buffer, sizeof(buffer));

    printf("Message from client: %s\n", buffer);

    close(client_fd);
    close(server_fd);

    unlink(SOCKET_PATH);

    return 0;
}
```

Output:

```
sharanaya@DESKTOP-TPK9V3Q:~/OSlab$    nano unix_socket_server.c
sharanaya@DESKTOP-TPK9V3Q:~/OSlab$    gcc unix_socket_server.c -o unix_socket_server
sharanaya@DESKTOP-TPK9V3Q:~/OSlab$    ./unix_socket_server
```

2. unix_socket_client.c

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main()
{
    int client_fd;
    struct sockaddr_un address;

    // Create socket
    client_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    // Set socket address
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    // Connect to server
    connect(client_fd,
            (struct sockaddr *)&address,
            sizeof(address));

    // Send message
    write(client_fd, "Hello Server", 12);

    printf("Message sent to server.\n");

    close(client_fd);

    return 0;
}
```

Output:

```
sharanaya@DESKTOP-TPK9V3Q:~/OSlab$    nano unix_socket_client.c
sharanaya@DESKTOP-TPK9V3Q:~/OSlab$    gcc unix_socket_client.c -o unix_socket_client
sharanaya@DESKTOP-TPK9V3Q:~/OSlab$    ./unix_socket_client

sharanaya@DESKTOP-TPK9V3Q:~/OSlab$
```